# Urban Cart Retail Shop Analytics

SQL Queries — Q1 to Q25 | PostgreSQL

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## Q1. How many total orders has UrbanCart received so far?

**SQL Query:**

```
SELECT
   COUNT(DISTINCT order_id) AS "Total Orders"
FROM public."FactOrderItems"
```

☐ *Result: 1,200 total orders*

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## Q2. Which cities generate the highest number of orders & revenue?

**SQL Query:**

```
SELECT
   c.city                          AS "City",
   COUNT(o.order_id)               AS "Total Orders",
   SUM(p.quantity * q.unit_price)  AS "Total Revenue"
FROM public."FactOrders" o
JOIN public."DimCustomers" c  ON o.customer_id = c.customer_id
JOIN public."FactOrderItems" p ON o.order_id   = p.order_id
JOIN public."DimProducts" q    ON p.product_id = q.product_id
GROUP BY 1
ORDER BY 2 DESC
```

☐ *Result: Barishal leads with 173 orders (14.4%) and highest revenue (14.18%)*

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## Q3. What percent of total customers uses Gmail?

**SQL Query:**

```
SELECT
   COUNT(CASE WHEN email ILIKE '%@gmail%' THEN 1 END) * 100
   / COUNT(customer_id)            AS "Gmail Users (%)"
FROM public."DimCustomers"
```

☐ *Result: 70% of customers use Gmail*

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## Q4. What is the monthly trend of total orders over time?

**SQL Query:**

```
SELECT
   TO_CHAR(order_date::DATE, 'Month YYYY')  AS "Month",
   COUNT(order_id)                          AS "Total Orders"
FROM public."FactOrders"
GROUP BY 1
ORDER BY 1 DESC
```

☐ *Result: Oct 2025 peaked at 413 orders. Dec 2025 dropped to 173.*

## Q5. What is the order completion rate across all orders?

**SQL Query:**

```
Select
Round ((count(case
when status ilike 'Completed' then 1 end)*100/Count(distinct order_id)),2) as "Completion Rate(%)"
from public."FactOrders"
```

☐ *Only 59.4% of orders are completed.*

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## Q6. What is the total revenue generated by UrbanCart?

**SQL Query:**

```
SELECT
   SUM(o.quantity * c.unit_price)  AS "Total Revenue (BDT)"
FROM public."FactOrderItems" o
JOIN public."DimProducts" c ON o.product_id = c.product_id
```

☐ *Result: Total Revenue = BDT 22,45,122*

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## Q7. Which product categories contribute the most to total revenue?

**SQL Query:**

```
SELECT
   c.category                    AS "Product Category",
   SUM(o.quantity * c.unit_price)   AS "Total Revenue"
FROM public."FactOrderItems" o
JOIN public."DimProducts" c ON o.product_id = c.product_id
GROUP BY 1
ORDER BY 2 DESC
```

☐ *Result: Fashion leads with BDT 513,550 in total revenue*

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## Q8. Which individual products generate the highest revenue?

**SQL Query:**

```
SELECT
   c.product_name                 AS "Product Name",
   SUM(o.quantity * c.unit_price)   AS "Total Revenue"
FROM public."FactOrderItems" o
JOIN public."DimProducts" c ON o.product_id = c.product_id
GROUP BY 1
ORDER BY 2 DESC
```

☐ *Result: Power Bank 10000mAh tops with BDT 304,000 in revenue*

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## Q9. What is the Average Order Value (AOV) and Average Basket Size?

### SQL Query — Average Order Value:

```
SELECT
  ROUND(
    SUM(o.quantity * c.unit_price) / COUNT(DISTINCT order_id), 2
  ) AS "AOV"
FROM public."FactOrderItems" o
JOIN public."DimProducts" c ON o.product_id = c.product_id
```

### SQL Query — Average Basket Size:

```
SELECT
  ROUND(
    SUM(quantity) / COUNT(DISTINCT order_id), 2
  ) AS "Average Basket Size"
FROM public."FactOrderItems"
```

☐ *Result: AOV = BDT 1,870.94 | Average Basket Size = 9.96 items*

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## Q10. Which products are at risk of stock-out due to high sales and low inventory?

### SQL Query:

```
SELECT
  product_name   AS "Product",
  category       AS "Product Category",
  stock          AS "Stock",
  CASE
    WHEN stock <= 100  THEN 'Critical'
    WHEN stock <= 200  THEN 'Low'
    WHEN stock <= 500  THEN 'Medium'
    ELSE 'Healthy'
  END            AS "Status"
FROM public."DimProducts"
ORDER BY 3 ASC
```

☐ *Result: Power Bank 10000mAh (stock=90) is Critical — top revenue product at restocking risk*

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## Q11. Which customers contribute the highest total revenue?

### SQL Query:

```
SELECT
  q.customer_id                  AS "Customer ID",
  r.full_name                    AS "Name",
  SUM(o.quantity * c.unit_price) AS "Total Revenue"
FROM public."FactOrderItems" o
JOIN public."DimProducts" c    ON o.product_id  = c.product_id
JOIN public."FactOrders" q     ON o.order_id    = q.order_id
JOIN public."DimCustomers" r   ON q.customer_id = r.customer_id
GROUP BY 1, 2
ORDER BY 3 DESC
LIMIT 10
```

☐ *Result: Top customer is Raisa (ID 70) with BDT 42,516 in total revenue*

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## Q12. Generate a list of cancellation rate by city; also the cancellation rate for customers.

### SQL Query — Cancellation Rate by City:

```
SELECT
  o.city                                          AS "City",
  COUNT(c.order_id)                               AS "Total Orders",
  COUNT(CASE WHEN c.status ILIKE '%cancelled%' THEN 1 END)      AS "Cancelled Orders",
  ROUND(
    COUNT(CASE WHEN c.status ILIKE '%cancelled%' THEN 1 END) * 100.0
    / COUNT(c.order_id), 2
  )                                               AS "Cancellation Rate (%)"
FROM public."FactOrders" c
JOIN public."DimCustomers" o ON c.customer_id = o.customer_id
GROUP BY o.city
ORDER BY "Cancellation Rate (%)" DESC
```

### SQL Query — Cancellation Rate by Customer:

```
SELECT
  o.full_name                                     AS "Customer Name",
  COUNT(c.order_id)                               AS "Total Orders",
  COUNT(CASE WHEN c.status ILIKE '%cancelled%' THEN 1 END)      AS "Cancelled Orders",
  ROUND(
    COUNT(CASE WHEN c.status ILIKE '%cancelled%' THEN 1 END) * 100.0
    / COUNT(c.order_id), 2
  )                                               AS "Cancellation Rate (%)"
FROM public."FactOrders" c
JOIN public."DimCustomers" o ON c.customer_id = o.customer_id
GROUP BY o.full_name
ORDER BY "Cancellation Rate (%)" DESC
```

☐ *Result: Highest city = Gazipur (27%). Highest customer = Tanvir (50%)*

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## Q13. Do male and female customers show different purchasing patterns by category?

### SQL Query:

```
SELECT
  c."Gender"                                      AS "Gender",
  COUNT(DISTINCT o.order_id)                      AS "Total Orders",
  SUM(p.unit_price * oi.quantity)                 AS "Total Revenue (BDT)",
  ROUND(AVG(p.unit_price * oi.quantity), 2)       AS "Avg Order Value",
  COUNT(CASE WHEN o.status ILIKE '%cancelled%' THEN 1 END)      AS "Cancelled Orders",
  ROUND(COUNT(CASE WHEN o.status ILIKE '%cancelled%' THEN 1 END)
     * 100.0 / COUNT(o.order_id), 2)              AS "Cancel Rate (%)"
FROM public."FactOrders" o
JOIN public."DimCustomers" c    ON o.customer_id  = c.customer_id
JOIN public."FactOrderItems" oi ON o.order_id     = oi.order_id
JOIN public."DimProducts" p     ON oi.product_id  = p.product_id
GROUP BY 1
```

☐ *Result: Male = BDT 1,300,780 (686 orders). Female = BDT 944,342 (514 orders). Cancel rate ~same*

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## Q14. How does customer purchasing behavior change over time since account creation?

### SQL Query:

```
SELECT
  dc.customer_id                                  AS "Customer ID",
  dc.full_name                                    AS "Customer Name",
  CASE
    WHEN (TO_DATE(fo.order_date, 'MM/DD/YYYY')
      - TO_DATE(dc.created_at,  'MM/DD/YYYY')) <= 30
      THEN '0-30 days'
```

```
    WHEN (TO_DATE(fo.order_date, 'MM/DD/YYYY')
      - TO_DATE(dc.created_at,  'MM/DD/YYYY')) <= 60
      THEN '31-60 days'
    ELSE '60+ days'
  END                                    AS "Period",
  COUNT(DISTINCT fo.order_id)            AS "Total Orders",
  SUM(foi.quantity)                      AS "Total Items"
FROM public."FactOrders" fo
JOIN public."DimCustomers" dc    ON fo.customer_id = dc.customer_id
JOIN public."FactOrderItems" foi ON fo.order_id    = foi.order_id
GROUP BY 1, 2, 3
ORDER BY 1, 2, 3
```

☐ *Result: Customer order ratios vary across time periods since account creation*

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## Q15. Generate a list of customers who ordered in October but didn't order in December.

### SQL Query:

```
SELECT DISTINCT
  c.customer_id   AS "Customer ID",
  c.full_name     AS "Customer Name",
  c.city          AS "Customer City"
FROM public."DimCustomers" c
WHERE c.customer_id IN (
  SELECT customer_id FROM public."FactOrders"
  WHERE EXTRACT(MONTH FROM TO_DATE(order_date, 'MM/DD/YYYY')) = 10
)
AND c.customer_id NOT IN (
  SELECT customer_id FROM public."FactOrders"
  WHERE EXTRACT(MONTH FROM TO_DATE(order_date, 'MM/DD/YYYY')) = 12
)
```

☐ *Result: 17 customers ordered in October but did not order in December*

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## Q16. Generate a list of customers who ordered both in October & December.

### SQL Query:

```
SELECT DISTINCT
  c.customer_id   AS "Customer ID",
  c.full_name     AS "Customer Name",
  c.city          AS "Customer City"
FROM public."DimCustomers" c
WHERE c.customer_id IN (
  SELECT customer_id FROM public."FactOrders"
  WHERE EXTRACT(MONTH FROM TO_DATE(order_date, 'MM/DD/YYYY')) = 10
)
AND c.customer_id IN (
  SELECT customer_id FROM public."FactOrders"
  WHERE EXTRACT(MONTH FROM TO_DATE(order_date, 'MM/DD/YYYY')) = 12
)
```

☐ *Result: 81 customers ordered in both October and December*

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## Q17. Which payment methods are used most frequently?

### SQL Query:

```
SELECT
```

```
    method                      AS "Payment Method",
    COUNT(DISTINCT order_id)        AS "Total Orders"
FROM public."FactPayment"
GROUP BY 1
ORDER BY 2 DESC
```

☐ *Result: COD = 488 | bKash = 349 | Nagad = 235 | Credit Card = 65 | Debit Card = 63*

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## Q18. Is there any relationship between payment method and order status?

**SQL Query:**

```
SELECT
    o.method                     AS "Payment Method",
    c.status                     AS "Order Status",
    COUNT(DISTINCT o.order_id)       AS "Total Orders"
FROM public."FactPayment" o
JOIN public."FactOrders" c ON o.order_id = c.order_id
GROUP BY 1, 2
ORDER BY 1, 3 DESC
```

☐ *Result: COD has highest completions (300). Debit Card has highest cancellation ratio (~29%)*

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## Q19. Do certain cities prefer specific payment methods?

**SQL Query:**

```
SELECT
    p.city                       AS "City",
    o.method                     AS "Payment Method",
    COUNT(DISTINCT o.payment_id)     AS "Transaction Count",
    ROUND(
        COUNT(DISTINCT o.payment_id) * 100.0
        / SUM(COUNT(DISTINCT o.payment_id)) OVER (PARTITION BY p.city, 2)
    )                            AS "City Share (%)"
FROM public."FactPayment" o
JOIN public."FactOrders" c     ON o.order_id    = c.order_id
JOIN public."DimCustomers" p   ON c.customer_id = p.customer_id
GROUP BY 1, 2
ORDER BY 1, 3 DESC
```

☐ *Result: COD is preferred in all cities. Dhaka is the only city where bKash leads*

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## Q20. Are higher-value orders associated with specific payment methods?

**SQL Query:**

```
SELECT
    o.method                     AS "Payment Method",
    SUM(p.quantity * q.unit_price)    AS "Total Revenue"
FROM public."FactPayment" o
JOIN public."FactOrderItems" p ON o.order_id    = p.order_id
JOIN public."DimProducts" q    ON p.product_id = q.product_id
GROUP BY 1
ORDER BY 2 DESC
```

☐ *Result: COD = BDT 898,761 | bKash = 649,221 | Nagad = 457,689 | Credit = 124,179 | Debit = 115,272*

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## Q21. What is the average number of items per order by payment method?

**SQL Query:**

```
SELECT
   c.method                    AS "Method",
   ROUND(AVG(o.quantity), 2)       AS "Average Number of Items"
FROM public."FactOrderItems" o
JOIN public."FactPayment" c ON o.order_id = c.order_id
GROUP BY 1
ORDER BY 2 DESC
```

☐ *Result: Debit Card leads with 2.63 avg items. bKash lowest at 2.46*

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## Q22. Create a list to show the difference between the category average price and product price.

**SQL Query:**

```
SELECT
   product_name                                         AS "Product Name",
   category                                             AS "Product Category",
   unit_price                                           AS "Unit Price",
   ROUND(AVG(unit_price) OVER (PARTITION BY category), 2)        AS "Category Avg Price",
   unit_price - ROUND(AVG(unit_price) OVER (PARTITION BY category), 2)
                                                      AS "Difference"
FROM public."DimProducts"
ORDER BY 5 ASC
```

☐ *Result: Earphones has highest negative difference (-350) vs category avg. Power Bank has highest positive*

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## Q23. Which product pairs are most frequently purchased together in the same order?

**SQL Query:**

```
SELECT
   f1.product_id   AS "Product 1",
   f2.product_id   AS "Product 2",
   COUNT(*)        AS "Pair Count"
FROM public."FactOrderItems" f1
JOIN public."FactOrderItems" f2
   ON  f1.order_id    = f2.order_id
   AND f1.product_id  < f2.product_id
GROUP BY 1, 2
ORDER BY 3 DESC
```

☐ *Result: Product 24 & 30 are the most frequently paired (22 times)*

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## Q24. Which product pairs generate the highest total revenue when sold together?

**SQL Query:**

```
SELECT
   f1.product_id   AS "Product 1",
   f2.product_id   AS "Product 2",
   SUM(
      (p1.quantity * p2.unit_price) + (p1.quantity * p2.unit_price)
   )            AS "Total Revenue"
FROM public."FactOrderItems" f1
JOIN public."FactOrderItems" f2
   ON  f1.order_id    = f2.order_id
```

```
    AND f1.product_id  < f2.product_id
JOIN public."FactOrderItems" p1 ON f1.product_id = p1.product_id
JOIN public."DimProducts" p2    ON f2.product_id = p2.product_id
GROUP BY 1, 2
ORDER BY 3 DESC
```

☐ *Result: Product 9 & 40 generate the highest paired revenue at BDT 11,369,600*

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## Q25. Generate a daily report: Date, Total Orders, Items, Completed, Cancelled, Revenue.

### SQL Query:

```
SELECT
    TO_DATE(o.order_date, 'MM/DD/YYYY')                    AS "Order Date",
    COUNT(DISTINCT o.order_id)                             AS "Total Orders",
    SUM(c.quantity)                                        AS "Total Items",
    COUNT(DISTINCT CASE WHEN o.status = 'Completed'
        THEN o.order_id END)                               AS "Completed Orders",
    COUNT(DISTINCT CASE WHEN o.status = 'Cancelled'
        THEN o.order_id END)                               AS "Cancelled Orders",
    COUNT(DISTINCT CASE WHEN o.status = 'Pending'
        THEN o.order_id END)                               AS "Pending Orders",
    SUM(c.quantity * q.unit_price)                         AS "Total Revenue"
FROM public."FactOrders" o
JOIN public."FactOrderItems" c ON o.order_id   = c.order_id
JOIN public."DimProducts" q    ON c.product_id = q.product_id
GROUP BY 1
ORDER BY 1
```

☐ *Result: Daily breakdown helps identify peak dates and fulfillment patterns over time*